Problem 13.6.1

Show that

a) $\sinh z = \sinh x \cos y + i \cosh x \sin y$

b) $\cosh z = \cosh x \cos y + i \sinh x \sin y$

Solution

a)

$$
\begin{aligned}
\sinh z &= \frac{1}{2}(e^z - e^{-z}) \\
&= \frac{1}{2}(e^{x+iy} - e^{-x-iy}) \\
&= \frac{1}{2}(e^x e^{iy} - e^{-x} e^{-iy}) \\
&= \frac{1}{2}(e^x(\cos y + i \sin y) - e^{-x}(\cos y - i \sin y)) \\
&= \frac{1}{2}(e^x \cos y + ie^x \sin y - e^{-x} \cos y + ie^{-x} \sin y) \\
&= \frac{1}{2}((e^x - e^{-x}) \cos y + i(e^x + e^{-x}) \sin y) \\
&= \boxed{\sinh x \cos y + i \cosh x \sin y}
\end{aligned}
$$

b)

$$
\begin{aligned}
\cosh z &= \frac{1}{2}(e^z + e^{-z}) \\
&= \frac{1}{2}(e^{x+iy} + e^{-x-iy}) \\
&= \frac{1}{2}(e^x e^{iy} + e^{-x} e^{-iy}) \\
&= \frac{1}{2}(e^x(\cos y + i \sin y) + e^{-x}(\cos y - i \sin y)) \\
&= \frac{1}{2}(e^x \cos y + ie^x \sin y + e^{-x} \cos y - ie^{-x} \sin y) \\
&= \frac{1}{2}((e^x + e^{-x}) \cos y + i(e^x - e^{-x}) \sin y) \\
&= \boxed{\cosh x \cos y + i \sinh x \sin y}
\end{aligned}
$$

Problem 13.6.3

Show that

a) $\cosh^2 z - \sinh^2 z = 1$

b) $\cosh^2 z + \sinh^2 z = \cosh 2z$

Solution

Problem 13.6.7

Find, in the form $u + iv$

a) $\cos i$

b) $\sin i$

Solution

Problem 13.6.13

Using the definitions, prove

a) $\cos z$ is even, $\cos(-z) = \cos z$
b) $\sin z$ is odd, $\sin(-z) = -\sin z$

Solution

Problem 13.6.16

Find all solutions

$$\sin z = 100$$

Solution

Problem 13.7.5

Find

$$\mathrm{Ln}(-11)$$

Solution

Problem 13.7.7

Find

$$\mathrm{Ln}(4 - 4i)$$

Solution

Problem 13.7.8

Find

$$\mathrm{Ln}(1 \pm i)$$

Solution

Problem 13.7.15

Find all values and graph some in the complex plane

$$\ln(e^i)$$

Solution

Problem 13.7.19

Solve for z

$$\ln z = 4 - 3i$$

Solution

Problem 13.7.23

Find the principal value.

$$(1+i)^{1-i}$$

Solution

Problem 14.1.1

Find and sketch the path

$$z(t) = (1 + \frac{1}{2}i)t \quad (2 \leqq t \leqq 5)$$

Solution

Problem 14.1.11

Find a parametric representation and sketch the path

$$\text{Segment from } (-1, 1) \text{ to } (1, 3)$$

Solution

Problem 14.1.19

Find a parametric representation and sketch the path

$$\text{Parabola } y = 1 - \frac{1}{4}x^2 \quad (-2 \leq x \leq 2)$$

Solution

Problem 14.1.21

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Re} z \; dz, \quad \text{C the shortest path from } 1+i \text{ to } 3+3i$$

Solution

Problem 14.1.25

$$\int_C z \exp(z^2)\ dz, \quad \text{C from 1 along the axes to } i$$

Solution

Problem 14.1.29

$$\int_C \operatorname{Im} z^2\ dz \quad \text{counterclockwise around the triangle with vertices } 0,\ 1,\ i$$

Solution